ZHE JIA, Ph.D.

AI Research Scientist • Foundation Models & Scalable Deep Learning

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Professional Summary

AI Research Scientist (PI) and Caltech Ph.D. with 8+ years architecting **Foundation Models**, **GenAI**, and **State Space Models**. Proven track record scaling autoregressive sequence modeling and bridging the Sim-to-Real gap. Author of 20+ top-tier papers (**NeurIPS, ICLR, Science, Nature Comms**). Expert in **Transformers** and **Distributed HPC**. Seeking roles to build and scale robust LLMs and GenAI architectures.

Technical Skills

GenAI & Architectures:	Transformers, State Space Models (Mamba), Latent Diffusion, Attention Mechanisms, FNO, RLHF / Alignment.
Systems & Distributed:	PyTorch, Distributed Parallel (Megatron/ZeRO), Memory-efficient Attention (vLLM/FlashAttention).
Languages & Tools:	Python, C++, CUDA (basic), Fortran, Bash, GitHub, CI/CD.

Professional Experience

Research Assistant Professor (Principal Investigator) — **2024 – Present**
The University of Texas at Austin — *Austin, TX*
Lead an AI lab bridging Foundation Models with physical systems. Secured 5M+ CPU/GPU-hours in computing grants.

- **Sensoformer (Set-Transformer & Sim-to-Real Alignment):** Architected early-fusion Set-Transformers for $O(1)$ global routing on dynamic tokens. Engineered *Physics-Structured Domain Randomization* and a continuous Focal L1 Loss to cure long-tail mode collapse. Achieved absolute **SOTA** without real labels, reducing latency by **1000x** (ICML 2026 under review).
- **Generative Physical AI (LDM & Neural Operators):** Coupled Sensoformer latent encoders with Latent Diffusion Models (LDM) and Fourier Neural Operators (FNO) for data-driven Bayesian inversion and physical super-resolution.
- **GTS (Autoregressive State Space Model):** Built the first dual-output Markovian neural surrogate for 10,000+ step rollouts over continuous state spaces in discipline. Solved exposure bias via Truncated BPTT and WSD LR scheduling. Leveraged PyTorch CUDA Graphs and adaptive time-stepping for 3x GPU acceleration.
- **Tech Leadership:** Managed AI roadmaps, mentored engineers to big tech roles, and built 10TB+ data pipelines.

Green Postdoctoral Fellow (HPC & Tech Lead) — **2022 – 2024**
Scripps Institution of Oceanography (UCSD) — *San Diego, CA*

- **HPC Simulation & Global Collaboration:** Led US-Turkey cross-functional team to resolve the mechanics of the 2023 Turkey earthquakes. Conducted dynamic simulation and high-dim inversions (C++/Fortran) on distributed clusters, discovering novel cascade rupture causes (**Science Cover**).
- **Automated ETL Pipeline:** Engineered a distributed pipeline for multi-TB data, extracting extreme long-tail signals (SNR < 0.1).

Graduate Research Assistant (Scientific Computing) — **2016 – 2022**
California Institute of Technology — *Pasadena, CA*

- **Deep Learning & Probabilistic Inversion:** Developed a CNN-UNet framework for signal processing (**100x speedup** over DSP); Built parallelized Bayesian (hybrid MCMC) inference engines for high-dimensional uncertainty quantification, overall reached 1000 citations.

Education

Ph.D. in Geophysics (Focus: Inverse Problems & Inference) — **2022**
California Institute of Technology (Caltech) — *Pasadena, CA*

B.S. & M.S. in Geophysics (Special Class for the Gifted Young - Top 1‰) — **2013 / 2016**
University of Science and Technology of China — *Hefei, China*

Selected Publications

*Selected from 20+ papers. * denotes corresponding or advisory author.*

- Liu, D., **Jia, Z.***, et al. (2026) *GTS: A Graph Transformer Foundation Model for Ultra-Long Autoregressive Dynamics.* **ICLR** (to be submitted). – *Bridging 15 Orders of magnitude with closed-Loop state tracking and stratified sampling.*
- **Jia, Z.***, et al. (2026) *Sensoformer: Robust Sim-to-Real Inference on Variable-Geometry Sensor Sets via Physics-Structured Randomization.* **NeurIPS** (in review).
- **Jia, Z.***, et al. (2023) *The complex dynamics of the Turkey earthquake doublet.* **Science** (Cover). – *Multi-modal distributed optimizations.*
- Zhang, X., **Jia, Z.***, et al. (2020) *Extracting dispersion curves from noise using deep learning.* **IEEE TGRS**. – *First CNN-UNet for seismic analysis, achieving 100x speedup over DSP.*

Selected Honors & Awards

Cecil H. and Ida M. Green Fellowship ($138k independent postdoc award) | **Caltech GPS Fellowship**	2022 | 2016
Outstanding Self-Financed Student Award (China National Award, Top 1‰)	2023
China National Scholarship for Graduate Students (Top 1% nationwide)	2015